Speaker Information

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Talk Information

Whole-Brain Intravital Imaging Reveals Asymmetric Serotonin-Dependent Post-Shock Firing in Drosophila

Abstract

Volumetric imaging of intact adult brains remains challenging due to optical scattering and the speed limitations of conventional scanning microscopy. Here we present V-shape Bessel-beam light-sheet microscopy (vSPIM), an upright, high-speed imaging platform that achieves subcellular resolution across large, opaque volumes in vivo. Compared to spinning-disk and confocal systems, vSPIM increases volumetric acquisition speed by over 30-fold while maintaining comparable signal quality and minimizing phototoxicity.

We applied vSPIM to perform real-time, whole-brain calcium imaging in adult Drosophila, uncovering anpreviously uncharacterized asymmetric, serotonin-dependent post-shock firing (PsF) response. This activity emerges within the mushroom body and dopaminergic neurons only after repetitive aversive stimulation and is distinct from canonical dopamine-driven reinforcement signals. vSPIM also resolved complex (18 distinct neural activity patterns) odor valence coding within the dense mushroom body calyx that aligns with connectomic architecture. Finally, we demonstrated the system's versatility by capturing rapid neuro-epithelial dynamics in the curved mouse cornea. vSPIM establishes a scalable framework for intravital imaging, bridging the gap between optical innovation and circuit physiology in intact adult tissues.

Research Overview (for collaboration discussion)

[Research Interests] Brain/tumor multiplex imaging and live imaging.

[Current Research Projects] High-speed Multiplex labeling in large FFPE biopsy /whole brain functional and super resolution imaging to study memory formation in Drosophila brain.

[Techniques] Tissue clearing, Multi-round organ antibody labeling, Lightsheet microscopy, Live calcium imaging

[Aspects or techniques of interest for future collaboration] Antibody/dye technologies